# Writing the methods section

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

## One complete research-case evidence record.

Specify exact inputs, boundaries, data roles and versions while keeping unassessed design details explicit.

**Keep one habit:** Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

### Figures that preserve evidence

**My remembered idea, example and correction**

### Robustness across conditions

**My remembered idea, example and correction**

### Reproducible environment and records

**My remembered idea, example and correction**

# W1 · Explain the current example

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- A methods section records question, units, sampling, permissions, data roles, baseline, procedure and preprocessing, metric, uncertainty assumptions, missing/exclusion handling, versions and deviations. Main report saying used AI on data and got good scores omits evidence needed to reconstruct the design and computation. Do not fabricate missing model or split information.

  Separate deterministic toy method applies Python string length to exact ASCII inputs oak, pine, willow without trimming. Counts three, four, six yield short/long/long under length at least four. A strictly-greater-than-four change yields short/short/long; pine is the boundary case. A trailing space changes oak to four characters because trimming is not part of the method.

  These are arbitrary string labels, not ecological categories or learned-model results. Reproducible toy output does not validate scientific sampling. Near-duplicate source grouping and split allocation must be documented; filenames alone do not establish source independence.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

# W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

## Complete task and evidence

- A methods section records question, units, sampling, permissions, data roles, baseline, procedure and preprocessing, metric, uncertainty assumptions, missing/exclusion handling, versions and deviations. Main report saying used AI on data and got good scores omits evidence needed to reconstruct the design and computation. Do not fabricate missing model or split information.

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- Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

| Main report omission | Why it affects assessment | What the audit should request |
|---|---|---|
| Model/procedure version | Which system produced the result is unknown | Exact version/configuration and code |
| Data identity and split | Cases, roles and source overlap unknown | Identifiers, eligibility and grouping/split rule |
| Baseline | Comparison target unknown | Baseline procedure and matched conditions |
| Metric and scoring | Good scores has no defined denominator/rubric | Explicit outcome, calculation and raw counts |

| Main report omission | Why it affects assessment | What the audit should request |
| --- | --- | --- |
| Missing/exclusion handling | Reported cases may be selected | Rules, excluded/unassessed counts and deviations |
| Validity conclusion | Cannot assess from this description | Record missing evidence rather than fabricate a complete method |

### My reasoning and evidence

MAIN LESSON · SECTION 7

## W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- A methods section records question, units, sampling, permissions, data roles, baseline, procedure and preprocessing, metric, uncertainty assumptions, missing/exclusion handling, versions and deviations. Main report saying used AI on data and got good scores omits evidence needed to reconstruct the design and computation. Do not fabricate missing model or split information.

  Separate deterministic toy method applies Python string length to exact ASCII inputs oak, pine, willow without trimming. Counts three, four, six yield short/long/long under length at least four. A strictly-greater-than-four change yields short/short/long; pine is the boundary case. A trailing space changes oak to four characters because trimming is not part of the method.

  These are arbitrary string labels, not ecological categories or learned-model results. Reproducible toy output does not validate scientific sampling. Near-duplicate source grouping and split allocation must be documented; filenames alone do not establish source independence.

- A report omits this qualification: A detailed **methods section** can reveal limitations; it does not guarantee a correct design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 9

# W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- A methods section records question, units, sampling, permissions, data roles, baseline, procedure and preprocessing, metric, uncertainty assumptions, missing/exclusion handling, versions and deviations. Main report saying used AI on data and got good scores omits evidence needed to reconstruct the design and computation. Do not fabricate missing model or split information.

  Separate deterministic toy method applies Python string length to exact ASCII inputs oak, pine, willow without trimming. Counts three, four, six yield short/long/long under length at least four. A strictly-greater-than-four change yields short/short/long; pine is the boundary case. A trailing space changes oak to four characters because trimming is not part of the method.

  These are arbitrary string labels, not ecological categories or learned-model results. Reproducible toy output does not validate scientific sampling. Near-duplicate source grouping and split allocation must be documented; filenames alone do not establish source independence.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

## Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

#### Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

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MAIN LESSON · SECTION 10

## Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

## Complete task and evidence

- Add a labelled writing the **methods section evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of writing the **methods section**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

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MAIN LESSON · SECTION 11

# Feedback · Repair and return

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |

| Evidence | My record |
| --- | --- |
| What I can now explain without notes | |
| One remaining question | |

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use W2: Trace the worked case for Writing the **methods section**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

# Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Use the worked case for Writing the **methods section**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

# Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

**My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

# Extra practice · Pictured case

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.
- **Input**s: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short. No whitespace trimming occurs.
- Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

### Method details decide the boundary case

| Exact input | Characters | Length ≥ 4 | Length > 4 |
|---|---|---|---|
| oak | 3 | short | short |
| pine | 4 | long | short ← changes |
| willow | 6 | long | long |

No whitespace trimming. The rule, input and inclusive/strict boundary must be recorded.
“Long” is a toy string label, not an ecological category or a learned-model result.

AI ACADEMY / AI-7/30

# Writing the methods section

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Record exact inputs

Preserve the three strings and no-trimming policy.



## 2. State procedure

Apply the inclusive length boundary at four characters.



## 3. Show resulting labels

Oak is short; pine and willow are long.

**CHANGE ONE THING**

Change the rule to length greater than 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

REFERENCE · SECTION 13

# Reference · Project milestone

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
| --- | --- |
| This week’s input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

# Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

## First supported question and optional hint

- Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.
- Which labelled step matches this detail? Oak is short; pine and willow are long.
- 1. State procedure
- 2. Record exact **input**s
- 3. Show resulting labels
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Show resulting labels Oak is short; pine and willow are long. A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring.

Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

## Changed-case question and optional hint

- Changed-case practice: Change the rule to length greater than 4. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Pine becomes short; willow stays long and oak stays short.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Pine becomes short; willow stays long and oak stays short. The equality case distinguishes the two procedures.

## Repair and retain the record

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

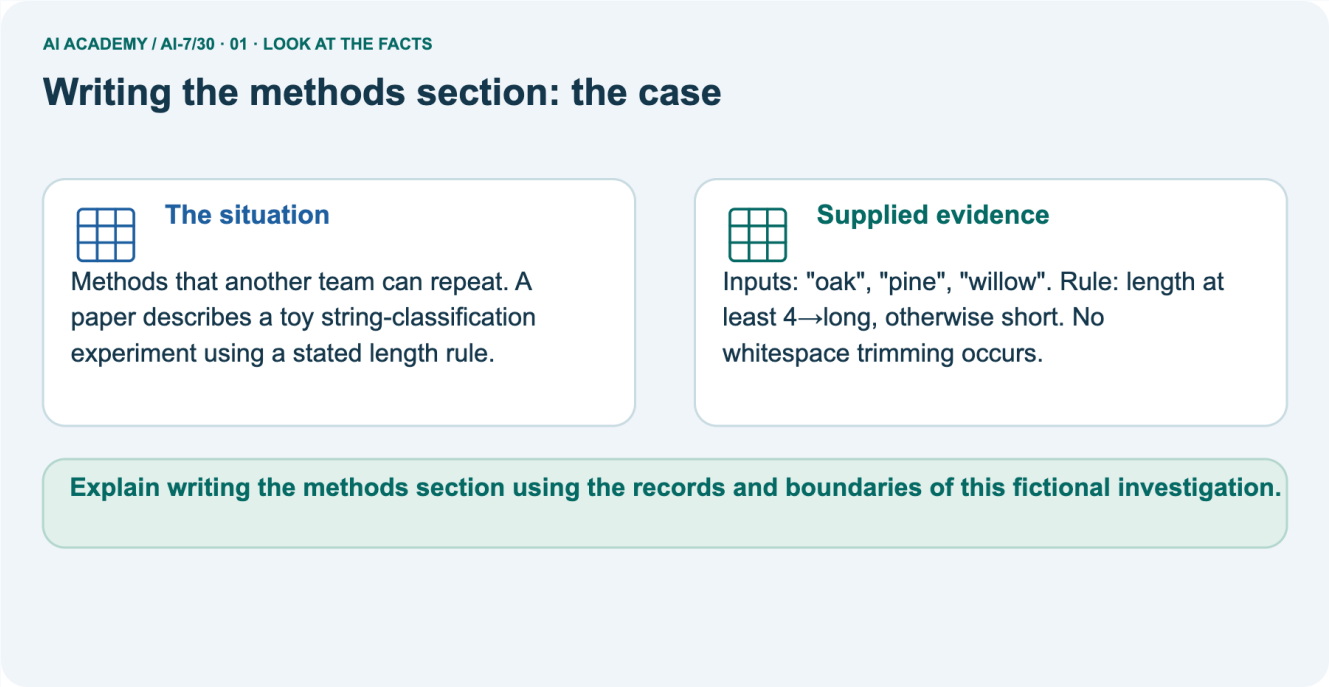
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

## Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

## 01. Writing the methods section: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

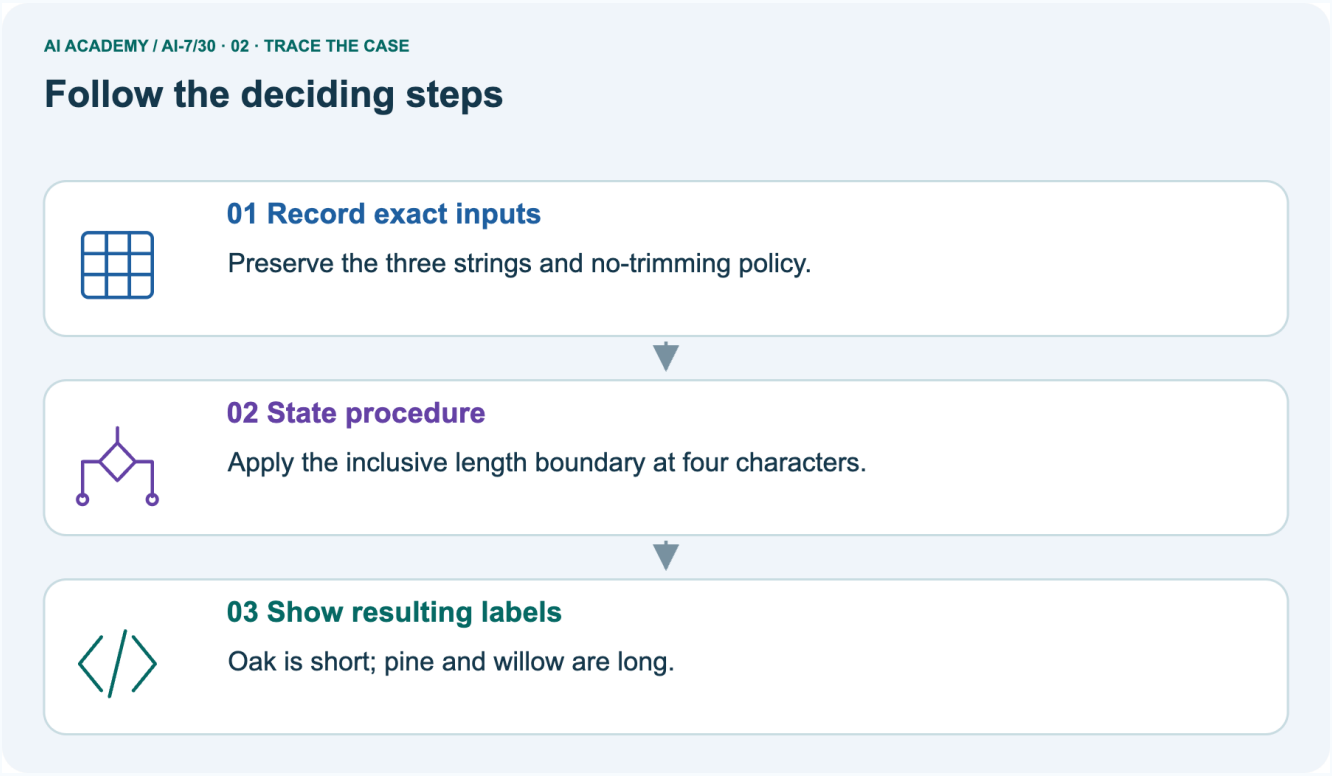
Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule. Supplied facts: **Input**s: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short.

No whitespace trimming occurs.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



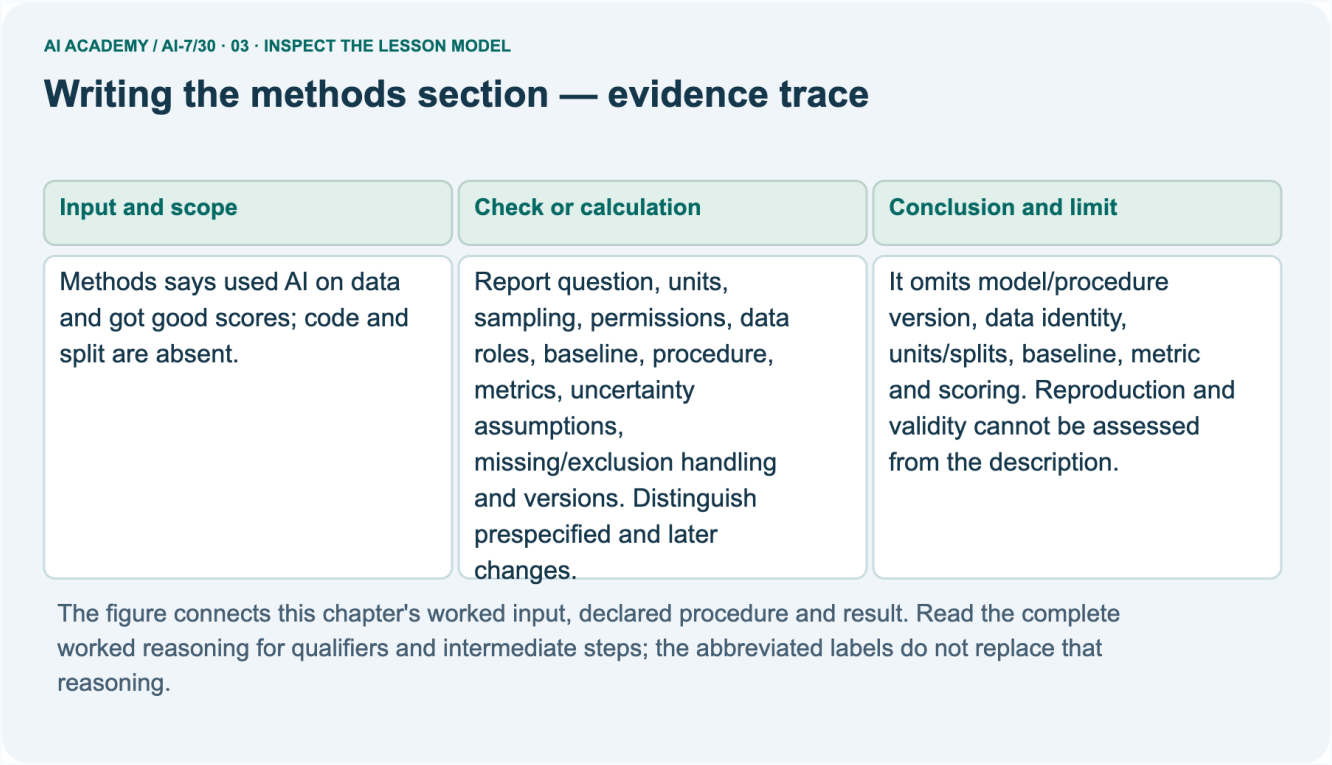

Follow the labelled picture with the matching explanation. Open the vector diagram

Record exact **input**s: Preserve the three strings and no-trimming policy. → State procedure: Apply the inclusive length boundary at four characters. → Show resulting labels: Oak is short; pine and willow are long.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Writing the methods section — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Writing the **methods section** — **evidence** trace.

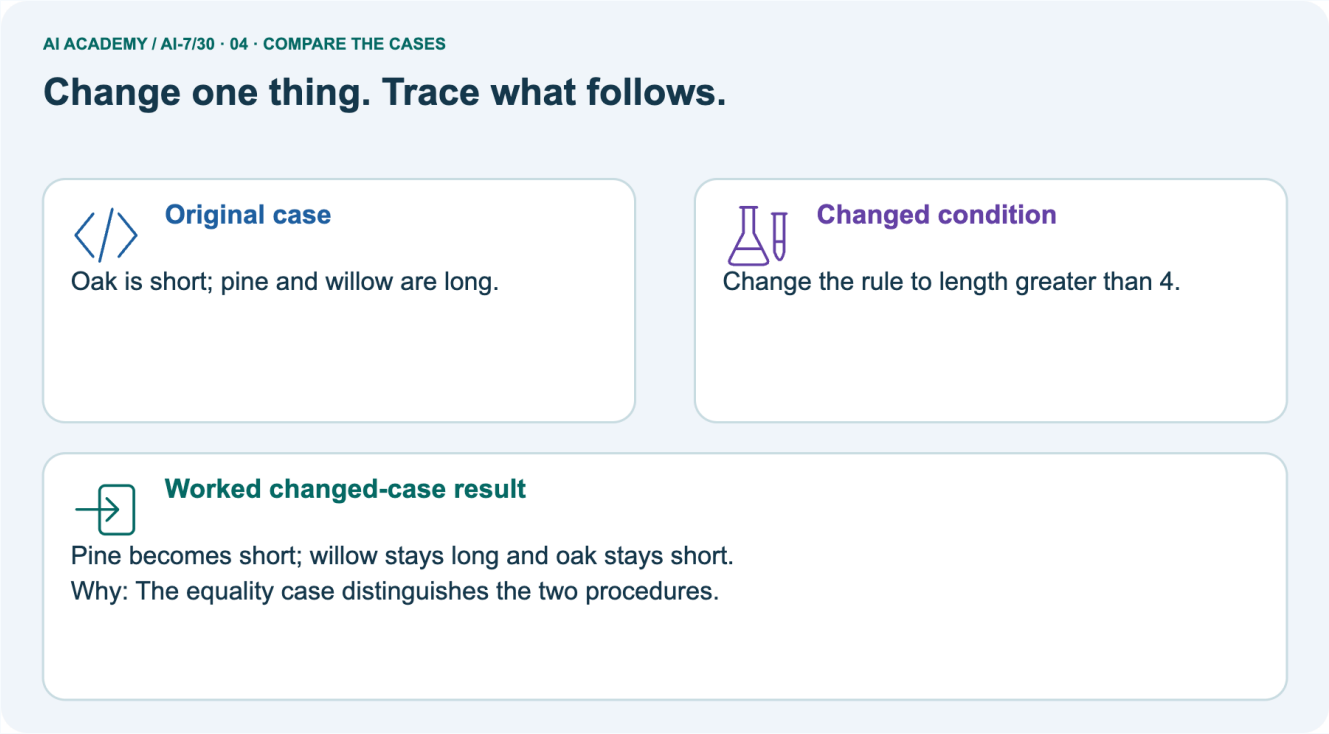
Methods says used AI on data and got good scores; code and split are absent. | Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions.

Distinguish prespecified and later changes. | It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

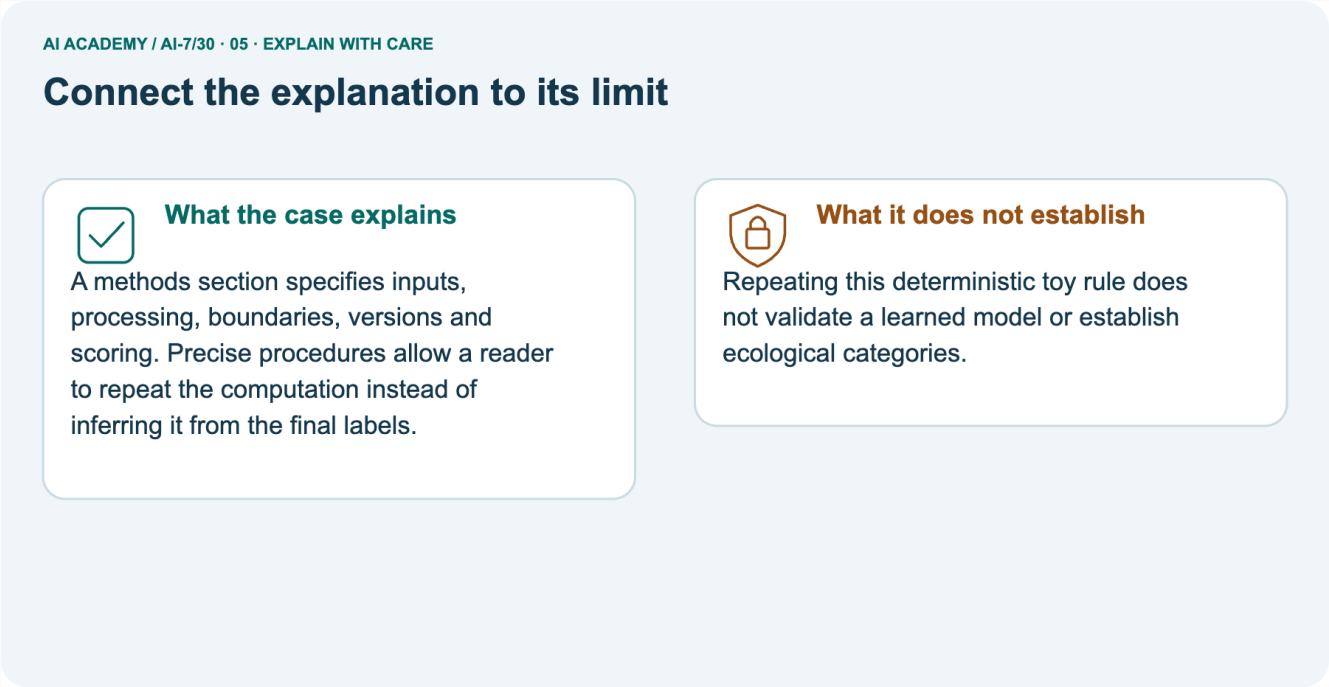
Original case: Oak is short; pine and willow are long. Changed condition: Change the rule to length greater than 4. Expected result: Pine becomes short; willow stays long and oak stays short.

Reason: The equality case distinguishes the two procedures.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

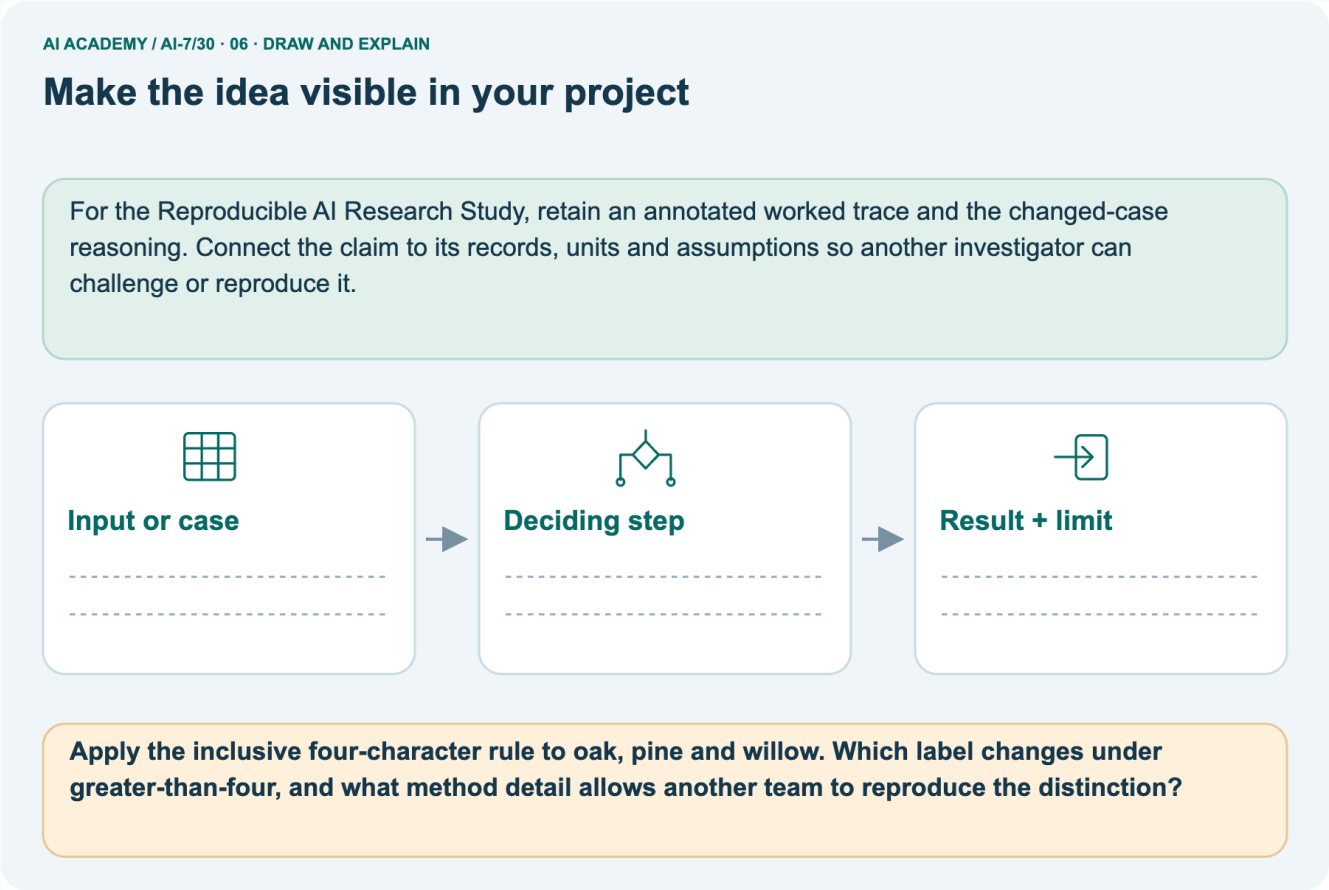
A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring. Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Limit: Repeating this deterministic toy rule does not validate a **learned model** or establish ecological categories.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

**01. Writing the methods section: the case**

**02. Follow the deciding steps**

**03. Writing the methods section — evidence trace**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.